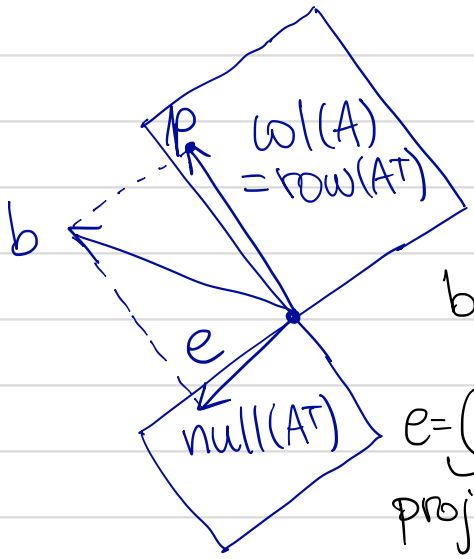
Lecture 24: Projections
Least squares
Best straight line

Proj. matrix $P = A(A^TA)^{-1}A^T$

If $\underbrace{b \text{ in } col(A)}_{b = Ax}$, $Pb = b$

If $\underbrace{b \perp col(A)}_{b \text{ in } null(A^T)}$, $Pb = 0$



$p = Pb$

$b = p + e$

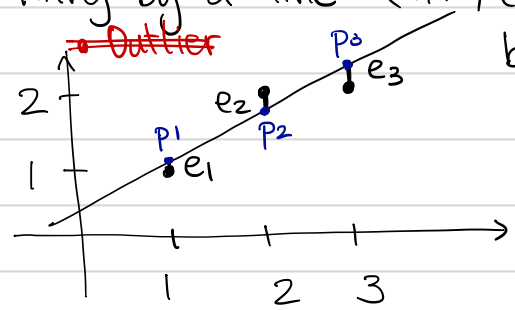
$e = \underbrace{(I-P)}_{\text{projects onto null}(A^T)} b$

Find the mistake:

$$\left.\begin{aligned} P &= A(A^TA)^{-1}A^T \\ &= A A^{-1}(A^T)^{-1}A^T \\ &= I \end{aligned}\right\} \text{Wrong unless } A^{-1} \text{ exists!}$$

Least squares

Fitting by a line $(1,1), (2,2), (3,2)$



$b = C + Dt$

$= \frac{2}{3} + \frac{1}{2}t$

$p_1 = \frac{7}{6}$

$p_2 = \frac{5}{3}$

$p_3 = \frac{13}{2}$

$$\begin{aligned} C + D &= 1 \\ C + 2D &= 2 \\ C + 3D &= 2 \end{aligned} \qquad \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$A \cdot x = b$$

Minimize: $\|Ax - b\|^2 = \|e\|^2$
$= e_1^2 + e_2^2 + e_3^2$

Find: $\hat{x} = \begin{bmatrix} \hat{C} \\ \hat{D} \end{bmatrix}$, $p = A\hat{x}$

$$\boxed{A^TA\hat{x} = A^Tb}$$ "normal equations"

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} = A^TA$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix} = A^Tb$$

$$\begin{cases} 3C + 6D = 5 \\ 6C + 14D = 11 \end{cases} \quad (*)$$

Calculus:

$$e_1^2 + e_2^2 + e_3^2 = (C+D-1)^2 + (C+2D-2)^2 + (C+3D-2)^2$$

$\hookrightarrow$ Setting (partials $=0$) also gives $(*)$

$\hookrightarrow$ $2D = 1 \Rightarrow \boxed{D = \frac{1}{2}}$

$3C = 5 - 3 = 2$

$\boxed{C = \frac{2}{3}}$

Best line: $b = C + Dt = \frac{2}{3} + \frac{1}{2}t$

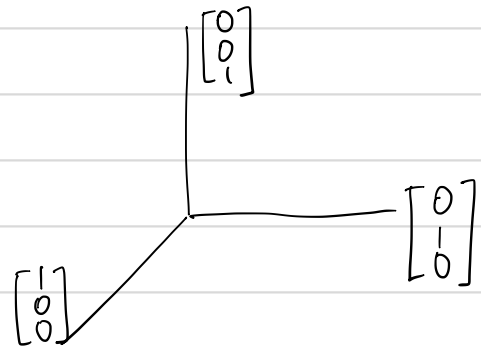
$$\underset{b}{\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}} = \underset{p}{\begin{bmatrix} 7/6 \\ 10/6 \\ 13/6 \end{bmatrix}} + \underset{e}{\begin{bmatrix} -1/6 \\ 2/6 \\ -1/6 \end{bmatrix}}$$

$p = A\hat{x}$ in col(A)

$e \perp$ col(A) in null(A^T)

Orthonormal vectors $q_1, q_2, \ldots$

$$q_i^T q_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq 0 \end{cases}$$



Fact: Orthonormal vectors are always independent.

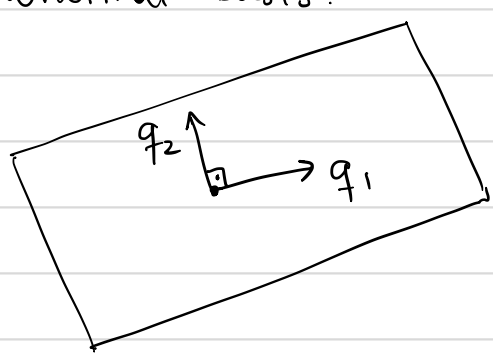
Proof:

$$q_1^T(a_1 q_1 + a_2 q_2 + a_3 q_3) = 0$$

$$\parallel$$

$$\underbrace{a_1 q_1^T q_1}_{a_1} + a_2 \cancel{q_1^T q_2}^{\,0} + a_3 \cancel{q_1^T q_3}^{\,0}$$

$\Rightarrow a_1 = 0$

Mult by q_2^T, q_3^T gives $a_2 = a_3 = 0$. ∎

Orthonormal basis: